Ridgelet-Nets With Speckle Reduction Regularization for SAR Image Scene Classification

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***Abstract*— With powerful feature representations, convolutional neural networks (CNNs) have produced tremendous achievements in image classification tasks and, typically, entail millions of labeled samples to train massive parameters. However, the sample labeling of synthetic aperture radar (SAR) images is extremely difficult, especially pixelwise labels, and has, sometimes, required field trips to accomplish labeling. Moreover, the inherent speckle noise may weaken the ability of networks to extract effective features from SAR images. In this article, we address these issues by labeling a few patchwise samples and propose Ridgelet-Nets with speckle reduction regularization for SAR image scene classification by combining deep learning with multiscale geometric analysis and statistical modeling of SAR images. First, we design Ridgelet-Nets with convolutional kernels constructed by ridgelet filters to reduce the training parameters and learn more discriminative features. Then, we embed speckle reduction regularization in the Ridgelet-Nets to restrain the influence of speckle noise and smooth the classification maps, in which the prior information of SAR image statistical modeling is introduced. Finally, we propose an adaptive SAR image scene classification framework based on an extended hierarchical visual semantic model, considering the differences in the structures and spatial relationships of different regions in the SAR images, particularly large-scale and complex scenes. Experimental results on real SAR images demonstrate that the proposed framework can achieve preferable classification performance using very limited labeled samples.**

***Index Terms*— Hierarchical visual semantic model, Ridgelet-Nets, scene classification, speckle reduction regularization, synthetic aperture radar (SAR) image.**

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I. Introduction

SYNTHETIC aperture radar (SAR) imaging systems [1], which can operate in all time and weather conditions, have been widely applied in many fields. Thus, a massive number of SAR images with large-scale and complex scenes are produced. Manually processing and analysis these images are becoming more and more difficult. It is necessary to develop effective automatic or semiautomatic SAR image understanding and interpretation algorithms [2]–[4]. Among them, the scene classification of SAR image [5], [6] in this article focuses on dividing diverse areas of an SAR image according to their functions into regions with the same semantics, such as residential, industrial, and commercial areas. Although these regions may all include buildings, roads, trees, and so on, the size, height, and spatial topological relations are different. Therefore, the intraclass variations and interclass similarities are more serious. In addition, the imaging mechanism of SAR images, the presence of inherent speckle noise [7], and the scene classification have become one of the most challenging tasks in SAR image interpretation.

Traditional classification methods [8], [9] are generally composed of feature extraction, followed by a trainable classifier, whose performance is highly reliant on the quality of handcrafted features, such as the features extracted by the gray-level co-occurrence matrix (GLCM) [10], the wavelet transform [11], the Gabor filtering [12], and multiscale geometric analysis [13]. Yu *et al.* [14] employed two kinds of discrete wavelet transforms (DWTs) to perform SAR sea ice classification, which verified the effectiveness of DWT in interpreting SAR image textures. Zhe *et al.* [15] constructed multiresolution and multidirectional features with contourlet coefficients and then utilized a minimum distance classifier for SAR image classification. Li *et al.* [16] extracted rich textural features by the ridgelet transform for SAR image segmentation and obtained satisfactory performance. These methods have achieved promising successes, especially for images with less complex scenes. Nevertheless, the two separate steps of feature extraction and classification cannot guarantee that the extracted features are optimal for classification.

Deep neural networks, such as convolutional neural networks (CNNs) [17]–[19], can learn hierarchical feature representations from raw data automatically, which are more discriminative and robust than handcrafted features. Therefore,

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many approaches based on CNNs have recently been proposed for SAR image classification [20], [21]. Zhao *et al.* [22] proposed a discriminant deep belief network for SAR image classification that was superior to traditional feature extraction-based and support vector machine (SVM) classification methods. Guo *et al.* [23] trained a relatively large CNN to classify SAR images that showed great potential for CNNs used in SAR image interpretation. Chen *et al.* [24] presented an all-convolutional network for SAR image target classification, which consists of sparsely connected layers without fully connected layers. The above image classification methods have achieved state-of-the-art performance with end-to-end architectures but require a tremendous amount of labeled samples to train the networks.

Although CNNs have great advantages in SAR image classification, there are still some challenges that deserve further study. First, most SAR images lack ground-truth information. Obtaining sufficient label information, especially pixelwise labels, is often a labor-intensive and time-consuming task, sometimes even requiring experts to perform field studies. In practical applications, it is relatively easy to label a few patchwise samples. Thus, our SAR image scene classification is a typical small-sample learning task. CNNs generally perform poorly and likely yield overfitting due to insufficient training of the parameters when they are used directly for small-sample learning problems. Huang *et al.* [25] proposed a transfer learning method that transfers knowledge learned from a large number of labeled source domain data to limited labeled target domain data, which can mitigate the negative effects introduced by the lack of sufficient labeled samples in the target domain. Second, due to the interference between backscatters, SAR images are affected by inherent speckle noise, which dramatically degrades the quality of SAR images and the subsequent classification performance. Jie *et al.* [26] adopted a graph cut-based spatial regularization after classification to attenuate the influence of speckle noise and yield superior classification performance.

Many strategies can effectively improve the above challenges. However, most of the existing methods do not consider the problems of insufficient training samples and poor quality (with speckle noise) at the same time. In this article, we construct an end-to-end architecture to jointly solve both of the problems. To deal with the small sample learning problem in SAR image scene classification, we propose Ridgelet-Nets, where each ridgelet kernel optimizes only three parameters. The number of network parameters is greatly reduced, and the demand for labeled samples is correspondingly decreased. The ridgelet kernels can also capture rich multiscale geometric properties [27] of terrains to enhance the feature representations. Moreover, we design two effective initialization schemes for the Ridgelet-Nets. Then, to keep critical information and restrain noise, we construct a speckle reduction regularizer, including a sparse reconstruction term and a statistical distribution [28] constraint, which is embedded in Ridgelet-Nets to weaken the speckle noise and further prevent the network from falling into overfitting. Unlike general sparse regularization [29], which focuses on only the parameters of networks without considering some prior information of data, our regularizer attempts to introduce the statistical information of SAR images to improve the robustness and classification performance of networks.

Generally, an SAR image is composed of homogeneous, heterogeneous, and extremely heterogeneous regions [30], which are the typical high-dimensional and heterogeneous characteristics. Therefore, it is difficult for different structural regions of an image to achieve the optimal result by using the same classification strategy. To alleviate these difficulties, Jie *et al.* [31] and Liu *et al.* [32] proposed a hierarchical visual semantic model. Liu *et al.* [33] employed this model for SAR image segmentation, which is mainly used in images with a single aggregated area. When dealing with large-scale and complex scene images, there are always different scenes gathering together. To meet new challenges, we extend the hierarchical visual semantic model to a high-level semantic space and construct a fused regional map to capture and pass on the aggregation differences so that the semantic information of primary, middle, and high levels can interact with each other.

Under the interaction of pixel and semantic spaces, an SAR image is divided into hybrid, structural, and homogeneous pixel subspaces with different structures. Then, we propose an adaptive SAR image scene classification framework for the three subspaces. In the hybrid pixel subspace, the bright and dark pixels change dramatically, and rich structural information, such as urban areas and forests, is aggregated together. To capture the structures as much as possible, we construct a Ridgelet-ResNet12 and a data augmentation strategy for the classification of this subspace. The structural pixel subspace includes the edges between different terrains, lines, and isolated targets, which only needs to be located and not classified. For the homogeneous pixel subspace, the changes in brightness within an area are small, such as farmland, water areas, and idle land. We design a Ridgelet-CNN5 with speckle reduction regularization to classify different scenes precisely. Finally, all the results are integrated to obtain the final scene classification map. The overall flowchart of the proposed SAR image scene classification framework is shown in Fig. 1. The main contributions of this article are summarized as follows.

1) To reduce the number of labeled samples and improve the classification performance, we design a general and lightweight convolution kernel, called ridgelet kernel. A ridgelet kernel only optimizes three parameters and can be easily embedded into any CNN to compose a Ridgelet-Net. In addition, we design two initialization schemes to accelerate the convergence speed of Ridgelet-Net.
2) To further advance the robustness of networks and classification performance, we design and embed a speckle reduction regularization, including the statistical distribution of SAR images into Ridgelet-Nets.
3) Considering the characteristics of SAR images with large-scale and complex scenes, we develop the hierarchical visual semantic model and construct an adaptive SAR image scene classification framework. The classification results achieve semantic consistency and

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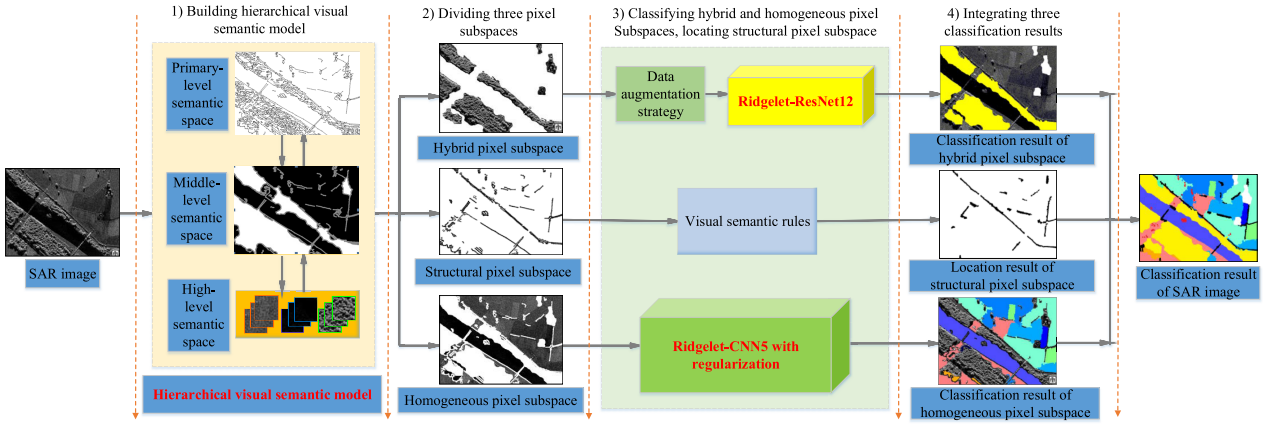


Fig. 1. Flowchart of the proposed SAR image scene classification framework.

structural preservation simultaneously, acquiring state-of-the-art performance.

The remainder of this article is organized as follows. Section II reviews some related works on ridgelet theory and the statistical distribution of SAR images. Then, in Section III, we describe the proposed Ridgelet-Nets, speckle reduction regularization, and the whole SAR scene classification algorithm in detail. Section IV presents and analyzes the experimental results. Finally, conclusions and future avenues of study are offered in Section V.

II. Related Work

In this section, we start with a brief review of the ridgelet theory and some related works. Then, we introduce the statistical distribution of SAR images.

A. Ridgelet Theory and Application

Human eyes are more sensitive to singularities than smooth regions of images. Similarly, the ridgelet [27], a multiscale wavelet, can capture and preserve the edge and linear singularities of images well. When the ridgelet is introduced into the representation of images, it contributes to learning more abundant directional information, especially for SAR images with rich textural and structural information.

In view of this characteristic, the ridgelet analysis has been widely used in many image processing fields. Yang *et al.* [34] proposed a ridgelet SVM for image noise reduction. Duan *et al.* [35] used a ridgelet kernel function in the local neighborhoods to preserve the details for SAR image segmentation. Fadili and luc Starck [36] summarized the ridgelet transforms and their applications in detail. These methods sufficiently explored the advantages of ridgelet theory and achieved outstanding results in various fields.

B. Statistical Distribution of SAR Images

Different from the common additive noise in optical images, the noise of SAR images, affected by the inherent imaging mechanism, is usually modeled by the multiplicative model [37]. It assumes that the SAR system return Λ is the product of two independent random fields: $\Lambda = X \cdot Y$, where X denotes the terrain backscatter that carries all the useful information from mapped areas and Y represents the speckle noise. Various distributions have been used for modeling the random fields X, Y, and Λ. Among them, Frery *et al.* [38] proposed the G^0 distribution to model the amplitude SAR images. They first introduced a reciprocal gamma law for the backscatter $X \sim \Gamma^{-1}(\sigma, \eta)$, which is characterized by the density

$$P_{\Gamma^{-1}}(X;\sigma,\eta) = \frac{\eta^{-\sigma}}{\Gamma(-\sigma)} \cdot X^{\sigma-1} \cdot \exp\left(-\frac{\eta}{X}\right) \quad \text{s.t. } -\sigma, \eta, X > 0 \tag{1}$$

where σ is related to the uniformity of terrain types; η reflects the average energy of the SAR image; and $\Gamma(x)$ denotes the gamma law.

They also defined the distribution of speckle noise Y: $Y \sim \Gamma(L, L)$ and derived the distribution of Λ: $\Lambda \sim G^0(\sigma, \eta, L)$; its density function can be written as

$$P_{G^0}(\Lambda;\sigma,\eta,L) = \frac{L^L \cdot \Gamma(L-\sigma) \cdot \Lambda^{L-1}}{\eta^\sigma \cdot \Gamma(L) \cdot \Gamma(-\sigma) \cdot (\eta + L \cdot \Lambda)^{L-\sigma}} \quad \text{s.t. } -\sigma, \eta, \Lambda > 0, L > 1 \tag{2}$$

where L represents the number of looks. When $L = 1$ (the speckle noise of an SAR image is the largest), the G^0 distribution can be expressed as follows:

$$P_{G^0}(\Lambda;\sigma,\eta) = \frac{-\sigma \cdot \eta^{-\sigma}}{(\eta + \Lambda)^{1-\sigma}} \quad \text{s.t. } -\sigma, \eta, \Lambda > 0, L > 1. \tag{3}$$

The G^0 law has the following advantages over other distributions: 1) it is a typical multiplicative model whose parameters are fully interpretable; 2) it is more tractable than many other multiplicative models from both the theoretical and computational viewpoints; and 3) it can model extremely heterogeneous, heterogeneous, and homogeneous areas well with different parameters. Therefore, the G^0 distribution is widely used in SAR image processing. Mejail *et al.* [39]

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estimated the parameters of the G^0 distribution as features for SAR image classification, and the results indicated that the parameters of G^0 distribution are interpretable and discriminable. Marques *et al.* [40] combined the G^0 distribution with the level set framework for SAR image segmentation and achieved great results.

III. Proposed Method

In this section, we introduce the proposed SAR image scene classification framework in detail. We first focus on the construction of ridgelet kernels and Ridgelet-Nets. Then, we describe the speckle reduction regularization, followed by the fusion of regional maps and the whole SAR image scene classification algorithm. Finally, we offer a complexity analysis of the proposed method.

A. Ridgelet Kernel

We introduce the ridgelet into CNNs and design ridgelet kernels to replace standard convolutional kernels, which can learn highly discriminative features and, more importantly, significantly reduce the parameters of networks.

Let (x, y) denotes the space domain of an image. For each scale $\alpha > 0$, orientation $\theta \in [0, 2\pi)$, and position $\beta \in R$, the continuous bivariate ridgelet function is defined as [36]

$$\psi_\gamma(x, y) = \alpha^{-1/2}\psi((x\cos\theta + y\sin\theta - \beta)/\alpha) \tag{4}$$

where $\gamma = \{\alpha, \theta, \beta\}$. The parameter α mainly affects the bandwidth, θ controls the direction, and β determines the position of ridgelets.

Let $Z = (x\cos\theta + y\sin\theta - \beta)/\alpha$; we utilize the difference of Gaussian (DoG) as the generating function of the ridgelet filter, which can be written as

$$\psi_r(Z) = \alpha^{-1/2}\left(e^{-\frac{Z^2}{2}} - \frac{1}{8}e^{-\frac{Z^2}{8}}\right). \tag{5}$$

In practical applications [41], the range of the parameters can be limited at certain intervals, $\alpha \in (0, 3)$, $\theta \in [0, \pi)$, and the range of β is related to θ, which is shown as follows:

$$\beta = \begin{cases} [0, N(\sin\theta + \cos\theta)], & \theta \in [0, \pi/2) \\ [N\cos\theta, N\sin\theta], & \theta \in [\pi/2, \pi) \end{cases} \tag{6}$$

where N represents the upper bound of the scale parameter. Fig. 2 shows several groups of 2-D-ridgelet filters constructed with different parameter values. We can clearly see the influence of the three parameters on the shape, direction, and position of ridgelet filters.

We design the ridgelet kernel according to (5). A ridgelet convolutional kernel with size $K \times K$ can be expressed as $\{(0, 0), (0, 1), \ldots, (k_i, k_j), \ldots, (K-1, K-1)\}$, where (k_i, k_j) denotes the position of the ridgelet kernel. $\psi_r(k_i, k_j)$ is used to calculate the value of the ridgelet kernel. Fig. 3 shows the calculation method of ridgelet kernels and the parameters comparison between ridgelet kernels and common convolutional kernels. We can see from Fig. 3 that the whole ridgelet kernel is modulated by the three parameters $\{\alpha, \theta, \beta\}$. That is, each ridgelet kernel optimizes only three parameters, while the common CNNs have to optimize 9 (3×3 kernel size) or 25 (5×5 kernel size) parameters per convolutional kernel.

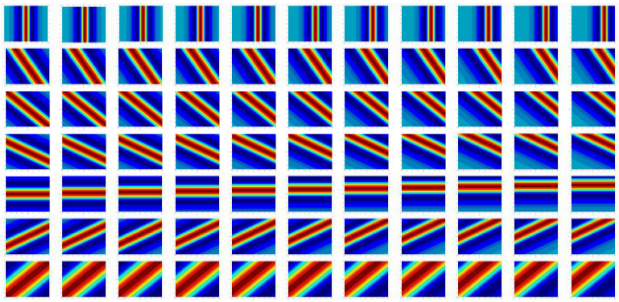


Fig. 2. Examples of 2-D-ridgelet filters. Each row has the same scale and orientation. Different rows have different orientations. The position parameter is different in each column, from 0 to 2 intervals of 0.2. The scale of the first row is 2, the last row is 0.5, and the others are 1.

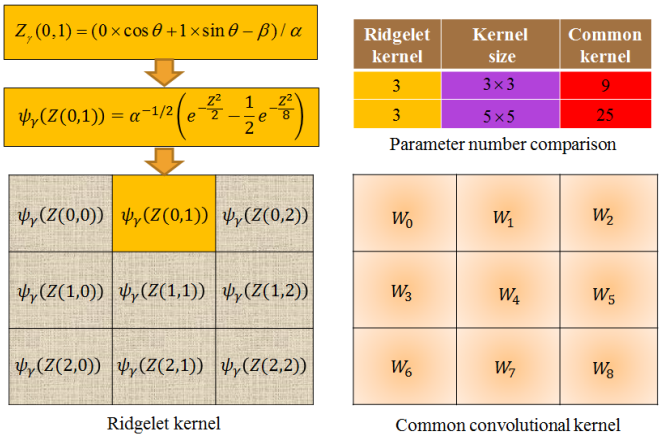


Fig. 3. Construction of the ridgelet kernel and comparison between the ridgelet kernel and the common convolutional kernel.

B. Ridgelet-Nets

It should be noted that the ridgelet kernel can be plugged into any deep convolutional network to construct a Ridgelet-Net. Meanwhile, the common elements or tricks in CNNs, such as pooling, dropout, skip connection, attention, adversarial, and activation functions, can be leveraged by the Ridgelet-Nets directly. Thus, we mainly introduce one of the convolutional layers in the Ridgelet-Net. The input feature maps in the lth convolutional layer are defined as

$$X^l = \left\{X_1^l, X_2^l, \ldots, X_{N_i}^l\right\} \tag{7}$$

where N_i denotes the number of input feature maps. The ridgelet kernels in the lth convolutional layer are defined as

$$F^l = \left\{F_1^l, F_2^l, \ldots, F_{N_o}^l\right\} \tag{8}$$

where $N_o = N_\alpha \times N_\theta \times N_\beta$ represents the number of channels or filters in the lth convolutional layer; N_α, N_θ, and N_β represent the number of scale, orientation, and position parameters in the lth convolutional layer, respectively.

The output feature maps in the lth convolutional layer are defined as

$$Y^l = \left\{Y_1^l, Y_2^l, \ldots, Y_{N_o}^l\right\} \tag{9}$$

where $Y^l = X^l * F^l$. For the first layer, X^l is the original SAR image patch; otherwise, $X^l = Y^{l-1}$.

Other structures of Ridgelet-Nets inherit common CNNs, such as LeNet, AlexNet, and ResNet. After building the

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network architecture, we introduce the initialization and update of the Ridgelet-Nets.

1) Initialization of Ridgelet-Nets: The initialization of Ridgelet-Net parameters is related to the number of channels in the network, which also determines the computational complexity of the network and the demand for labeled samples. An appropriate number of channels can also relieve the overfitting problem. Thus, we design two initialization schemes for Ridgelet-Nets.

To maintain the diversity of learned features as much as possible, we follow the empirical settings used for handcrafted ridgelet feature extraction and provide a simple and effective uniform initialization strategy. We use $\{\alpha_0^l, \theta_0^l, \beta_0^l\}$ to denote the initialization parameters of the ridgelet kernel in the lth convolutional layer, where $\alpha_0^l = \{\alpha_{0,1}^l, \ldots, \alpha_{0,i}^l, \ldots, \alpha_{0,N_\alpha}^l\}$, $\alpha_{0,i}^l$ is obtained by uniform sampling from $(0, 3)$. $\theta_0^l = \{\theta_{0,1}^l, \ldots, \theta_{0,i}^l, \ldots, \theta_{0,N_\alpha}^l\}$, $\theta_{0,i}^l$ is obtained by uniform sampling from $[0, \pi)$, $\beta_0^l = \{\beta_{0,1}^l, \ldots, \beta_{0,i}^l, \ldots, \beta_{0,N_\alpha}^l\}$, and $\beta_{0,i}^l$ is obtained by uniform sampling according to formula (6).

Sketch line segments contain the scale and orientation information of different terrains, which are crucial to capture the image features. Therefore, we propose a sketch statistics-based initialization method. We carry out histogram statistics for the length and orientation of the sketch line segments in the sketch map. For initialization of the scale parameters, we count the length of sketch line segments in the hybrid pixel subspace, normalize the length to $(0, 3)$, and calculate the proportion of the number of line segments of different lengths. The length of top N_α is used to initialize the parameters α_0^l. To initialize the orientation parameters, we first divide $[0, \pi)$ into 18 intervals to represent 18 orientations. Then, we count the number of orientations of all sketch line segments in the hybrid pixel subspace and calculate the proportion of sketch line segments falling in the 18 orientations. We take the orientations of the top N_θ to initialize the parameters θ_0^l. The initialization of β_0^l is the same as the first method. Fig. 4 shows the statistical results of the length and orientation of the sketch map of an SAR image.

2) Updating of Ridgelet-Nets: The main purpose of the Ridgelet-Nets is to reduce the number of parameters to be updated in the backpropagation. Instead of updating the weight value of each position in the convolutional kernels, we update the ridgelet kernels by adjusting the three ridgelet parameters. According to (5), we can derive the gradients of the ridgelet parameters. The gradient of the loss function L with respect to the scale parameter α is as follows. For simplicity, we omit subscripts and variables in the following formulas:

$$\frac{\partial L}{\partial \alpha} = \frac{\partial L}{\partial F} \cdot \frac{\partial F}{\partial \psi} \cdot \frac{\partial \psi}{\partial \alpha} = \sum_{x,y} \delta_F \alpha^{-\frac{5}{2}} Z \left(e^{-\frac{Z^2}{2}} - \frac{1}{8} e^{-\frac{Z^2}{8}} \right) (x \cos\theta + y \sin\theta - \beta). \tag{10}$$

The gradient of the loss function L with respect to the orientation parameter θ is

$$\frac{\partial L}{\partial \theta} = \sum_{x,y} \delta_F \alpha^{-\frac{3}{2}} Z \left(e^{-\frac{Z^2}{2}} - \frac{1}{8} e^{-\frac{Z^2}{8}} \right) (x \sin\theta - y \cos\theta). \tag{11}$$

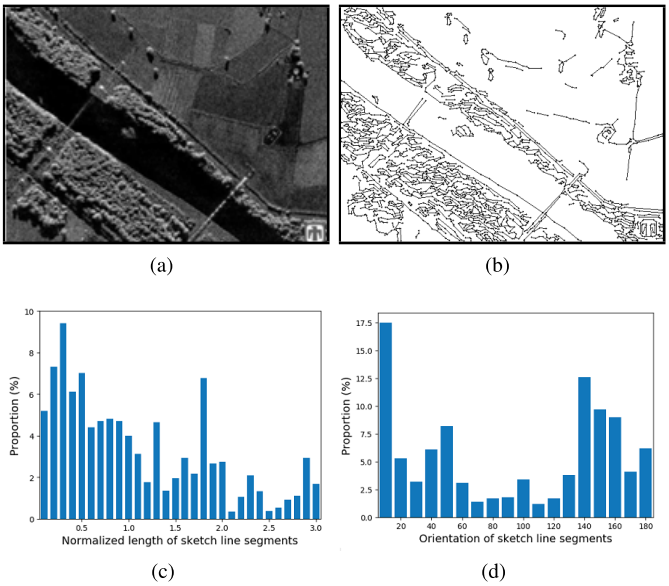


Fig. 4. Statistical results of an SAR image. (a) SAR image. (b) Sketch map. (c) Length statistical histogram. (d) Orientation statistical histogram.

The gradient of the loss function L with respect to the orientation parameter β is

$$\frac{\partial L}{\partial \beta} = \sum_{x,y} \delta_F \alpha^{-\frac{3}{2}} Z \left(e^{-\frac{Z^2}{2}} - \frac{1}{8} e^{-\frac{Z^2}{8}} \right) \tag{12}$$

where δ_F denotes the gradient of the loss function L with respect to F [42].

C. Speckle Reduction Regularization

We attempt to weaken the influence of speckle noise on the classification performance and further relieve overfitting by introducing regularization. Thus, we construct a regularizer named speckle reduction regularization in this article.

The detailed network structure of Ridgelet-Net with speckle reduction regularization is shown in Fig. 5. It mainly consists of an encoder, a decoder, and several fully connected layers. Following the variational autoencoder (VAE) [43], the fully connected layers are mainly used to estimate the parameters of G^0 and Γ^{-1} distribution. These parameters are used to calculate the regularization. Then, the total loss function is obtained by adding the regularization term to the classification loss, as follows:

$$L_{all}(x, l; \omega) = \lambda L_c(x, l; \omega_c) + (1 - \lambda) L_r(x; \omega_r, \omega) \tag{13}$$

where x represents the input SAR image patch, l denotes the label of samples, $L_c(\cdot)$ is the classification loss, $L_r(\cdot)$ denotes the speckle reduction regularization, $\omega = \{\omega_c, \omega_r\}$ indicates corresponding model parameters, and λ is a hyperparameter that determines the proportion of $L_c(\cdot)$ and $L_r(\cdot)$ in the total loss.

The speckle reduction regularization includes three items: a reconstruction error item that is mainly used to reduce the irrelevant components from the feature dimension; a symmetrical Kullback–Leibler (KL) distance constraint that can suppress noise components from the data distribution; and

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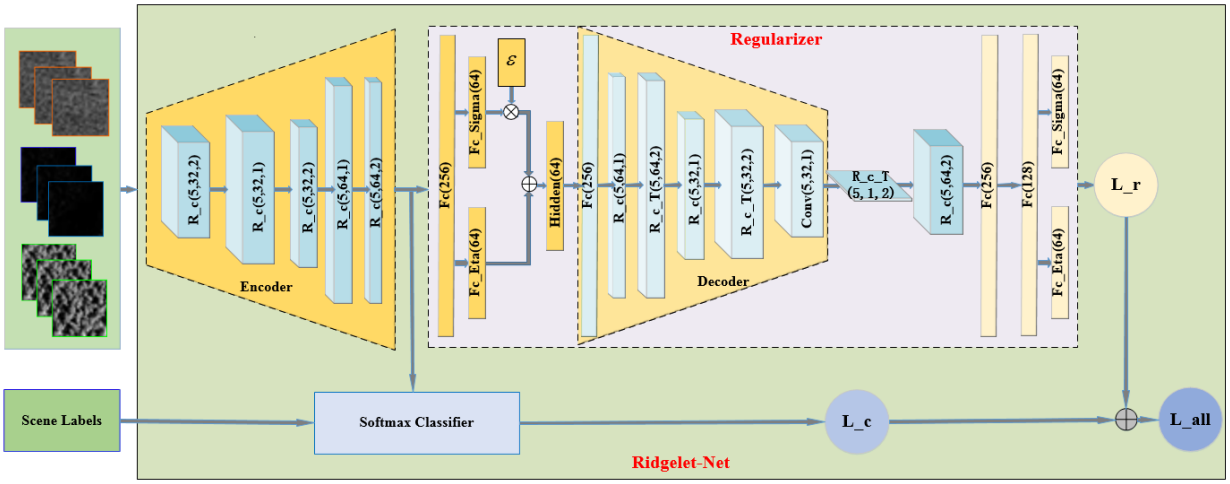


Fig. 5. Architecture of the Ridgelet-Net with speckle reduction regularization, where R_c(5,32,2) denotes ridgelet convolutional layer parameters (kernel size, filter number, and stride); Fc_Sigma(64) indicates that the output of the fully connected layer with 64 nodes is the estimated value of σ; $\varepsilon \sim N(0, 1)$; R_c_T denotes transposed ridgelet convolutional operation; and $\otimes$ and $\oplus$ represent the elementwise multiplication and sum operations, respectively.

an L_2-norm item to decrease the whole model complexity. We empirically demonstrate that the three terms together can obtain the optimal classification performance, which can be written as

$$L_r(x; \omega_r, \omega) = \sum_N \|x - \hat{x}\|^2 + D_{KL}(x, \hat{x}) + \|\omega\|_2 \quad (14)$$

where $\hat{x}$ represents the reconstructed image, N is the number of samples, and $D_{KL}(x, \hat{x})$ denotes the KL distance between two different distributional estimations; one is the distribution of raw SAR images $p_{G^0}(x)$, and the other is the distribution of useful terrain backscatters $q_{\Gamma^{-1}}(\hat{x})$, where $q_{\Gamma^{-1}}$ denotes the estimation of reciprocal gamma law. The KL distance is defined as follows [30]:

$$D_{KL}(x, \hat{x}) = \frac{1}{2} \sum_N (p_{G^0}(x) - q_{\Gamma^{-1}}(\hat{x})) \log \frac{p_{G^0}(x)}{q_{\Gamma^{-1}}(\hat{x})}. \quad (15)$$

The regularization is mainly used in the homogeneous pixel subspace. For the hybrid pixel subspace, especially in high-resolution SAR images, an image patch usually contains multiple targets; thus, its distribution may exhibit overdispersion or more than one mode. The authors' opinion is that it is more accurate to model the hybrid pixel subspace with mixed distributions than what can be explained by the G^0 model, which will be our future work.

D. Fusion of Regional Maps and Adaptive Scene Classification Algorithm for SAR Images

The existing hierarchical visual semantic model consists of primary- and middle-level semantic spaces without introducing high-level semantic information, so it has not been used in SAR image classification. Moreover, the aggregated regions in the middle-level semantic space are calculated by a single aggregated degree threshold T and K-nearest neighbor [33], where T determines which sketch line segments are candidate segments, and K decides whether the candidate segments are bilaterally aggregated. However, with the increases in image scale and scene complexity, the structural information and spatial relationships in different aggregated regions are notably different, and the single aggregated region cannot represent the aggregated characteristics of an image well. Thus, we merge regional maps with different aggregated degree thresholds and K-nearest neighbors to capture the differences between different areas.

From the observation of SAR images used in the experiments, it can be seen that the aggregated regions usually consist of several different scenes. For example, in the W. DC image [see Fig. 6(a)], residential and commercial scenes are aggregated together. In the residential scene, as shown in the red square, buildings are low, trees can be seen, and there is almost no shadow. The purple square exhibits the commercial area, where the buildings are very high and form many shadows. Both scenes generate rich sketch lines, as shown in Fig. 6(b), but the aggregated degree is different. The differences should be captured and inherited when calculating the regional map. We first calculate the aggregation statistical histogram under different K-nearest neighbors. For $K = 7$, the aggregated degree threshold is $T = 10.7$, which is calculated based on the peak value of the aggregation statistical histogram, as shown in Fig. 6(c), and the red line indicates the position of the threshold. For $K = 15$, the peak value of the aggregated degree is larger than $K = 7$, and the aggregated threshold is also different, as shown in Fig. 6(d). After that, based on different thresholds, the obtained aggregated areas of regional maps are very different, as shown in Fig. 6(e) and (f). To inherit and pass on these differences, we fuse the two regional maps with different aggregated degree thresholds and K-nearest neighbors. The fused regional map can provide effective guidance for the manual labeling of categories, as shown in Fig. 6(g), where white and gray areas represent the aggregated region and black areas are the homogeneous region. At the same time, we expand the hierarchical visual semantic spaces into three levels and introduce the semantic labels as the high-level semantic space so that the semantic information

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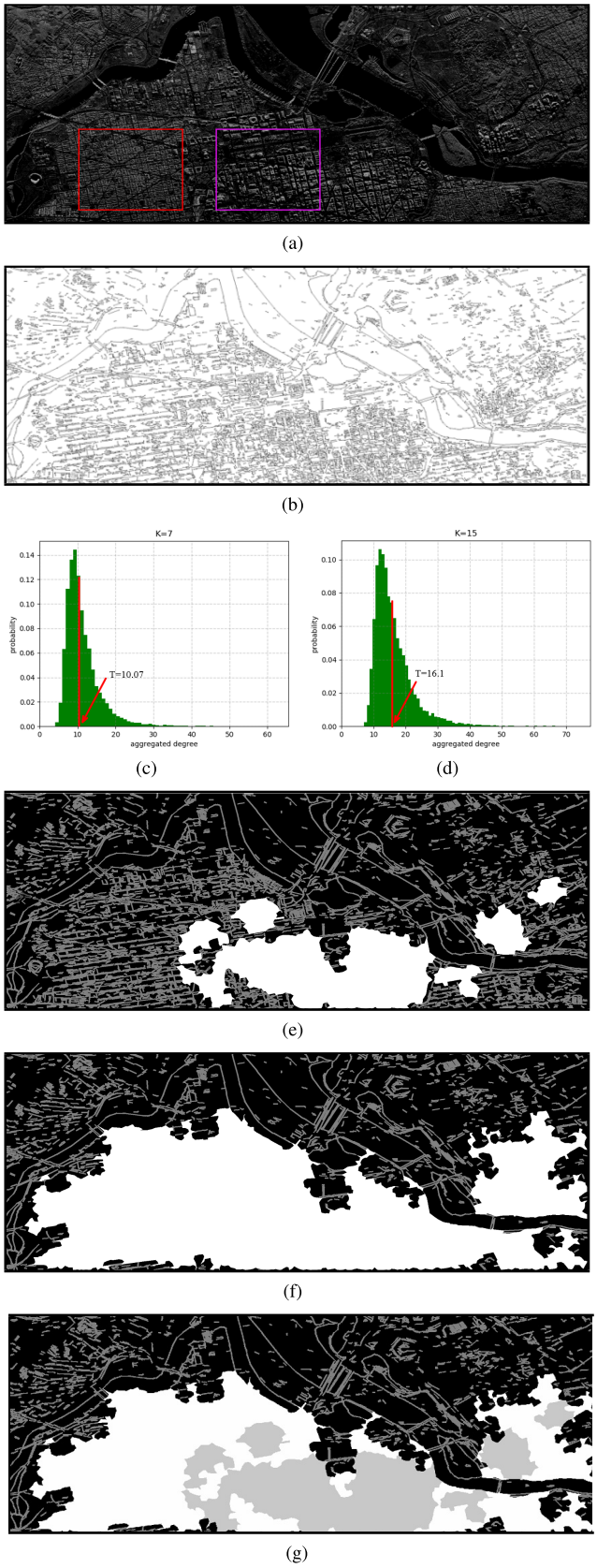


Fig. 6. Fusion of regional maps. (a) Original SAR image. (b) Sketch map. (c) Statistical histogram of aggregated degree ($K = 7$). (c) Statistical histogram of aggregated degree ($K = 15$). (e) Regional map ($K = 7$). (f) Regional map ($K = 15$). (e) Fused regional map, where white and gray represent different aggregated areas, respectively.

of primary, middle, and high levels can interact with each other and can be further utilized by the scene classification task.

For SAR image scene classification, it is better to adopt differentiated approaches to classify an SAR image according to the properties of different regions. Thus, based on the extended hierarchical visual semantic model, we design an adaptive SAR image scene classification algorithm. This is a notable difference from the existing supervised classification methods. First, we calculate the sketch map and fused the regional map of an SAR image successively. Under the guidance of the regional map, three-pixel subspaces with different structures can be obtained by mapping the regional map back to the original SAR image. After that, we employ patchwise methods to classify hybrid and homogeneous pixel subspaces. To obtain labeled samples for training classifiers, we manually label some image blocks under the guidance of fused regional maps. Since the terrain types in the hybrid pixel subspace are the most complex, to derive more structural information and keep the scattering characteristics of the ground objects, we design a Ridgelet-ResNet12 and a data augmentation strategy for classification of the hybrid pixel subspace. The data augmentation strategy includes magnifying, shrinking, and rotating the labeled sample block. The homogeneous pixel subspace of an SAR image is usually the largest, but the structural information is relatively less, so we instantiate a Ridgelet-CNN5 with speckle reduction regularization (Ridgelet-CNN5-SR) to distinguish different scenes accurately. The structural pixel subspace mainly includes edges, line targets, and isolated targets. We use some visual semantic rules [33] to locate them, such as roads and bridges. Finally, all the results are integrated together to obtain the final classification map. The detailed algorithm description is listed in Algorithm 1.

E. Computational Complexity of the Ridgelet-Nets

In this section, we investigate the time and space complexity of Ridgelet-Nets. The Ridgelet-Nets are mainly composed of convolutional, pooling, and fully connected layers. Generally, the pooling and connected layers occupy only 5%–10% of the computational complexity in a network [44]. Hence, we mainly analyze the computational complexity of the convolutional layers. There are two steps in the calculation of the ridgelet convolution layers. The first step is to calculate the ridgelet kernels using the ridgelet function, and the second step is the ordinary convolutional operation.

The total time complexity is the sum of the time complexity of all convolution layers, as follows:

$$O\left(\sum_{l=1}^{L} 3F_l^2 \cdot C_{l-1} \cdot C_l + K_l^2 \cdot F_l\right) \tag{16}$$

where L is the number of convolutional layers, l represents the index of the convolutional layer, F_l denotes the spatial size of the output feature map of the lth layer, C_{l-1} is the number of input channels in the $l-1$th layer, C_l is also known as the number of input channels of the lth layer, and K_l is the size of the filters.

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Algorithm 1 Adaptive SAR Image Scene Classification Method

Input: SAR image patches and related parameters;
Output: Classification result of SAR image;

1: Divide subspaces. Based on the extended hierarchical visual semantic model, an SAR image is divided into hybrid, homogeneous and structural pixel subspaces.
2: Extract training samples. Manually label a few image blocks for each scene.
   2.1 For the hybrid pixel subspace, each block is scaled and rotated, and then 100 small patches are randomly sampled on each block;
   2.2 For the homogeneous pixel subspace, randomly sample 200 small patches on each block directly.
3: Classify the hybrid pixel subspace.
   3.1 Initialize the Ridgelet-ResNet12;
   3.2 Train the network with the enhanced training samples;
   3.3 Obtain the classification result of the hybrid pixel subspace.
4: Classify the homogeneous pixel subspace.
   4.1 Initialize the Ridgelet-CNN5-SR;
   4.2 Train the network with the total loss function (13);
   4.3 Obtain the classification map of the homogeneous pixel subspace.
5: Locate the structural pixel subspace. Based on some visual semantic rules to locate line objects, such as roads, bridges.
6: Integrate the three results together to obtain the final classification map.

Space complexity, strictly speaking, includes two parts: the total number of parameters and the sum of output feature maps, which is calculated as follows:

$$O\left(\sum_{l=1}^{L} 3C_{l-1}\cdot C_l + \sum_{l=1}^{L} F_l^2\cdot C_l\right). \tag{17}$$

From the above analysis, we can see that the computational complexity of the Ridgelet-Net is much lower than that of common CNNs.

IV. Experiments and Analysis

In this section, we evaluate the proposed method on real SAR images. First, related experimental settings are introduced. After that, we provide qualitative and quantitative experimental results, followed by parameter analysis and discussion. Finally, we give the ablation study of the proposed method.

A. SAR Images and Related Settings

We utilize four high-resolution SAR amplitude images with large-scale and complex scenes, where the amplitude range of the last two images is $0 \sim 65\,535$, and other details are listed in Table I. There are no public annotations for these images, so we manually label an image block with a size of 100×100 for each scene. In the homogeneous pixel subspace, we randomly sample 200 patches with a size of 16×16 on each block as training samples. In the hybrid pixel subspace, we perform multiscale operations to increase the number of blocks by 21 times. Each block is magnified and minimized two times and then rotated by seven angles $\{45°, 90°, 135°, 180°, 225°, 270°, 315°\}$. After that, we randomly sample 100 patches with size 24×24 on each block as training samples.

We instantiate two Ridgelet-Nets in this article; Ridgelet-CNN5 consists of five convolutional layers with 5×5 ridgelet kernels, where the number of filters in the first three layers is 32 ($N_\alpha = 4, N_\theta = 4, N_\beta = 2$) and is doubled ($N_\alpha = 4, N_\theta = 8, N_\beta = 2$) in the last two layers, and the structure mainly follows the AlexNet. Ridgelet-ResNet12 contains four residual blocks, and each block has three convolutional layers with 3×3 ridgelet kernels. The number of filters starts from 32 and is doubled every next block. Both networks use two fully connected layers as classifiers. We choose the sketch statistics-based initialization method to initialize the two networks. The hyperparameter λ is set as 0.8.

To compare the performance of the proposed method, we design several related comparative experiments, including traditional supervised classifiers with handcrafted features and state-of-the-art CNN networks, as follows.

1) *GLCM-MLR:* We employ the GLCM to extract 16-D textural features for each patch and a multinomial logistic regression (MLR) for classification.
2) *Gabor-SVM:* The 24-D Gabor features and an SVM are applied to perform the patchwise classification task.
3) *CNN5-SR:* We embed the speckle reduction regularization term into the AlexNet for classification.
4) *ResNet12:* We use the ResNet12 directly for SAR image scene classification.

B. Qualitative Analysis of SAR Images

We use four SAR images [see Figs. 7–10(a)] to evaluate the effectiveness of the proposed method. In addition, we downloaded the corresponding optical images from Google Earth (Google Inc.) for reference [see Figs. 7–10(b)]. We also provide sketch maps [see Figs. 7–10(c)] and fused regional maps [see Figs. 7–10(d)] of SAR images.

The aggregated regions of the Bridge image are relatively simple, so we provide only the regional map of a single aggregated degree. The classification maps are shown in Fig. 7, where different colors represent the water (blue), forest (green), idle land (light yellow), and residential (magenta) scenes. There are many misclassifications in GLCM-MLR, as shown in Fig. 7(e), particularly in water and forest scenes. Fig. 7(f) illustrates the result of the Gabor-SVM method, where the classification accuracy of idle land and water scenes is improved compared with that of GLCM-MLR. The regional consistency of idle land and water scenes in Fig. 7(g) is better than that of Gabor-SVM and GLCM-MLR. The better performance indicates that the learned features are more effective than the handcrafted ones. However, there are still some misclassifications in CNN5-SR, and some shadows of the forest are mistakenly divided into water. Fig. 7(h) shows the classification map obtained by ResNet12. Except for the water scene, the classification accuracy of ResNet12 [see Fig. 7(h)] is

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TABLE I
DETAILED INFORMATION ON SAR IMAGES

Image	Size	Resolution	Band	Modality	Location	Sensor	Scene category
Bridge	350×901	$1\ m$	*X*-band	Spotlight	New Mexico	Airborne SAR	4
Noerdlin	3543×1506	$1\ m$	*X*-band	Spotlight	Swabian Jura	TerraSAR	5
W. DC	2000×810	$1\ m$	*Ku*-band	Stripmap	Washington D.C	Airborne SAR	5
Pyram	4134×3196	$1\ m$	*X*-band	Spotlight	Giza	TerraSAR	4

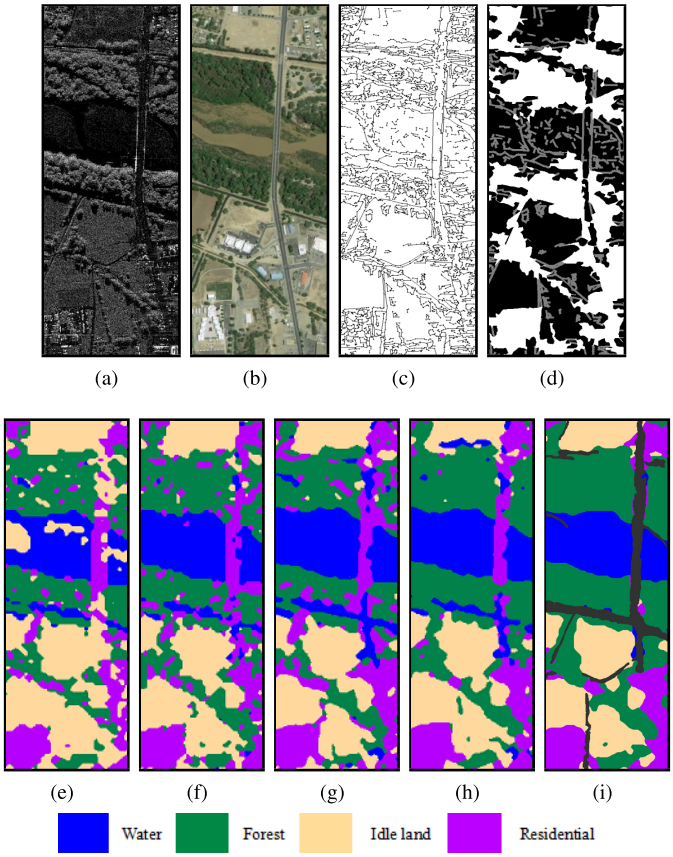


Fig. 7. Classification maps of the Bridge image. (a) Original SAR image, 350×901, *X*-band, and 1-*m* resolution. (b) Optical image. (c) Sketch map, which has three parameters: code length gain and high and low thresholds $(15, 1.2, 0.7)$. (d) Regional map ($K = 7$). (e) GLCM-MLR. (f) Gabor-SVM. (g) CNN5-SR. (h) ResNet12. (i) **Our method**.

further improved. It is obvious that our method [see Fig. 7(i)] yields the most accurate result, and the classification of water is more exact than the comparative approaches. This is because the design of Ridgelet-Nets can successfully avoid the overfitting issue associated with very limited labeled samples. In addition, we try to locate some line targets according to the visual semantic rules [33] and information interactions between the pixel and semantic spaces, and we mark them in black color, as shown in Fig. 7(i).

The fused regional map of the Noerdlin image is shown in Fig. 8(d), which divides the aggregated regions into two categories represented in white and gray. Thus, we label the aggregated regions with two types of scenes. The classification maps are presented in Fig. 8(e)–(i), where different colors denote the farmland (blue), grassland (green), idle land (red), industrial (yellow), and residential (magenta) scenes. The result of the GLCM-MLR is shown in Fig. 8(e); we can see that it fails to differentiate the five scenes, and there are severe misclassifications. The classification result of Gabor-SVM [see Fig. 8(f)] outperforms that of GLCM-MLR, but the regional connectivity is poor and presents small granular appearances due to the existence of speckle noise. These results indicate that the handcrafted features are not satisfactory for all scenes of SAR images. The classification performance and spatial connectivity of the CNN5-SR method are obviously superior to the result of Gabor-SVM, as shown in Fig. 8(g), which indicates that the CNNs have powerful feature representations, and the introduction of speckle reduction regularization can effectively weaken the noise. However, the classification result is still not satisfying, and many boundaries are misclassified into industrial scenes. The classification result of ResNet12 [see Fig. 8(h)] is more accurate than that of CNN5-SR; especially, the misclassifications of boundaries are decreased greatly. Our method yields the best classification result and achieves semantic consistency and structural preservation simultaneously compared with the comparative methods, as shown in Fig. 8(i), and the located line targets are marked in white color. This is because the proposed method adopts the adaptive scene classification framework for different pixel subspaces. Meanwhile, we design Ridgelet-Nets and speckle reduction regularization, resulting in more discriminative and robust features and better performance for SAR image scene classification.

We can see from the SAR image of W. DC [see Fig. 9(a)] and its sketch map [see Fig. 9(c)] that the aggregated degrees of the building on the left and right of the image are different. Therefore, we fuse two regional maps of different aggregated degrees, and the fused result is shown in Fig. 9(d), which is a more accurate and diversified representation of the aggregation of terrain types than the regional map with a single aggregated degree. Then, we manually annotate an image block in each aggregated area as the training area. The classification maps of the W. DC image are shown in Fig. 9(e)–(i), where different colors denote the water (blue), grassland (green), idle land (yellow), residential (red), and commercial (purple) scenes. The results of GLCM-MLR [see Fig. 9(e)] and Gabor-SVM [see Fig. 9(f)] are worse than the CNN-based methods, which are difficult to distinguish different scenes except for waters. It can be seen that the generalization ability of traditional methods is very poor when labeled samples are limited, especially for large-scale and complex scene images. The CNN5-SR achieves superior performance over handcrafted feature methods, as shown in Fig. 9(g). The classification of residential is more accurate, and the regional consistency is improved but performs worse in the water scene because waters and shadows

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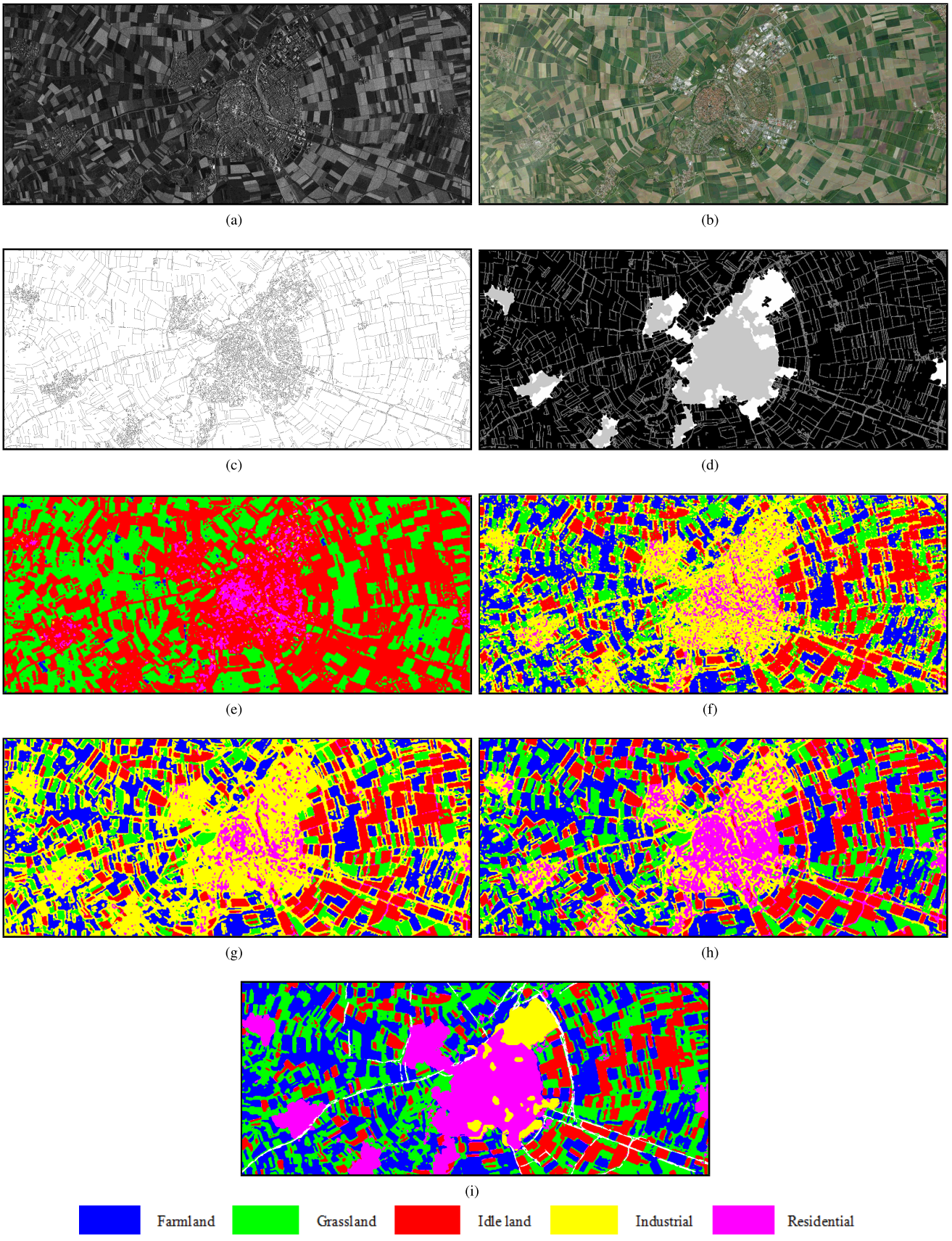


Fig. 8. Classification maps of the Noerdlin image. (a) Original SAR image, 3543 × 1506, *X*-band, and 1-*m* resolution. (b) Optical image. (c) Sketch map (25, 1.5, 0.8). (d) Fused regional map, $K = 5$ and 13. (e) GLCM-MLR. (f) Gabor-SVM. (g) CNN5-SR. (h) ResNet12. (i) **Our method**.

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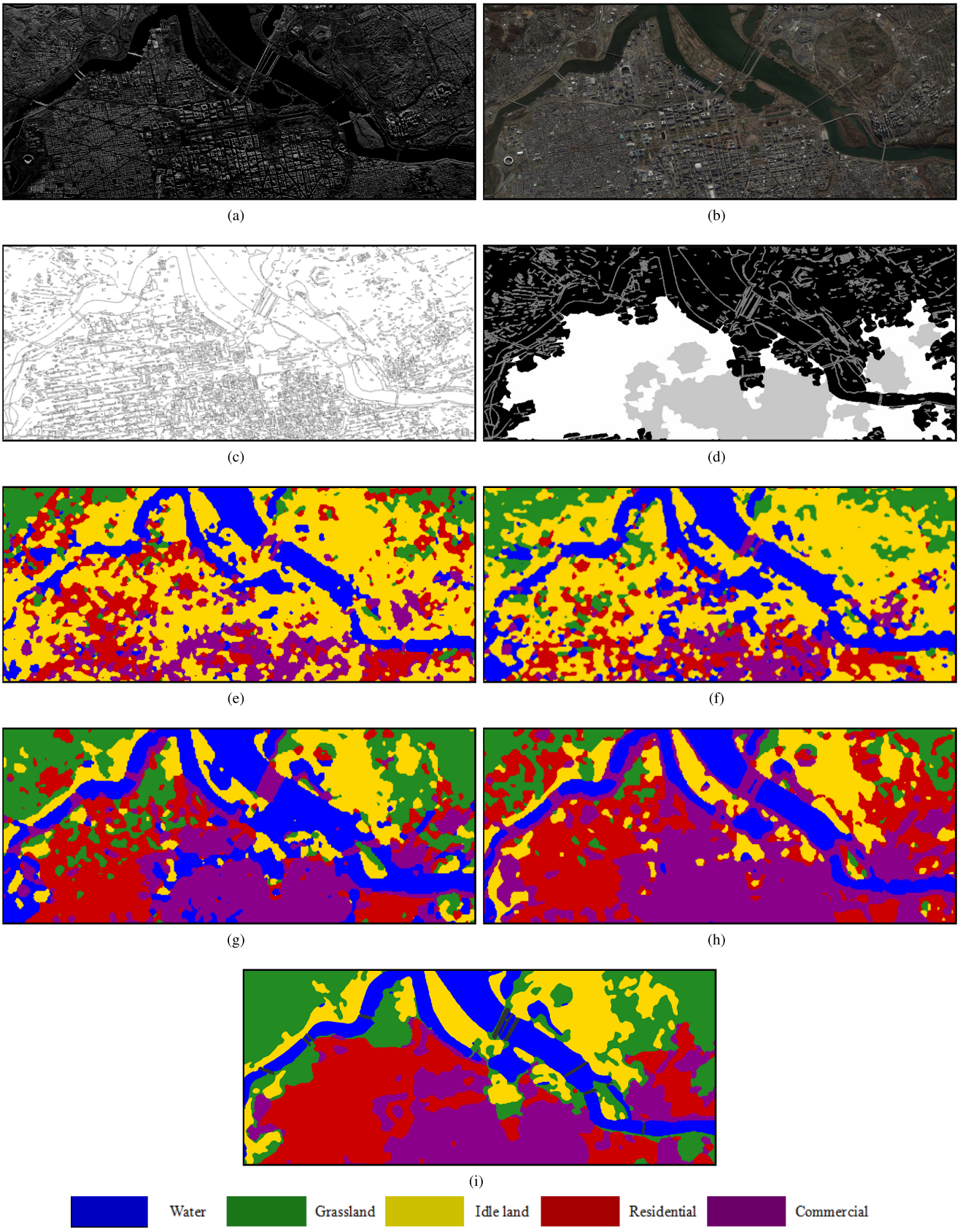


Fig. 9. Classification maps of the W. DC image. (a) Original SAR image, 2000 × 810, *Ku*-band, and 1-*m* resolution. (b) Optical image. (c) Sketch map (5, 0.3, 0.1). (d) Fused regional map, $K = 7$ and 15. (e) GLCM-MLR. (f) Gabor-SVM. (g) CNN5-SR. (h) ResNet12. (i) **Our method**.

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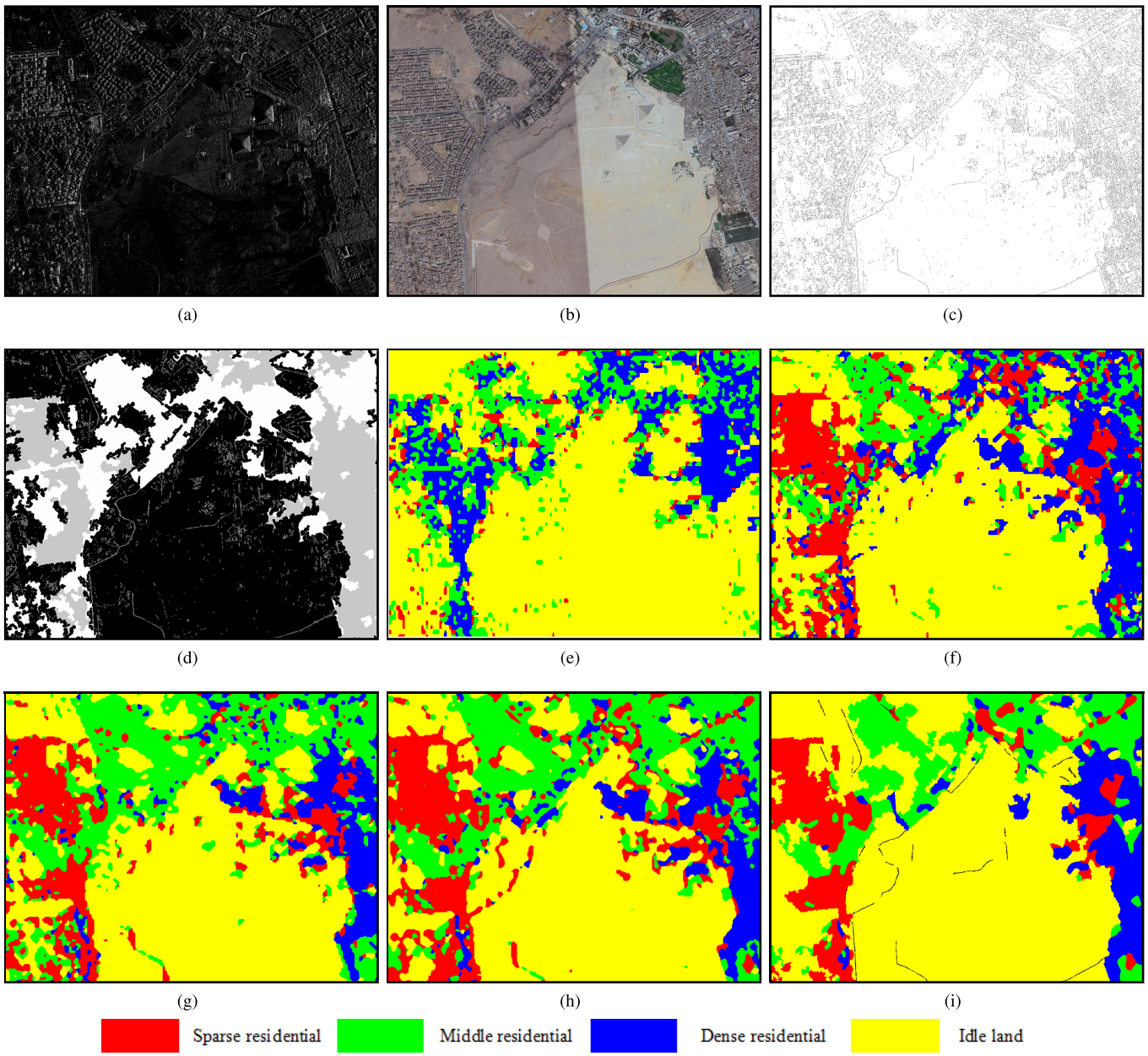
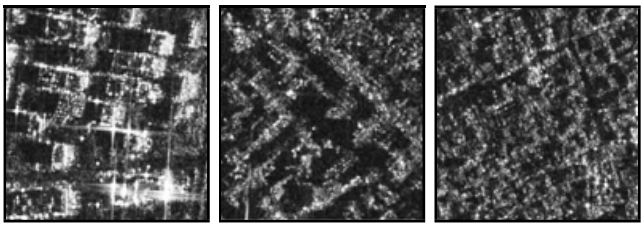


Fig. 10. Classification maps of the Pyram image. (a) Original SAR image, 4134 × 3196, *X*-band, and 1-*m* resolution. (b) Optical image. (c) Sketch map (5, 0.2, 0.1). (d) Fused regional map, $K = 7$ and 15. (e) GLCM-MLR. (f) Gabor-SVM. (g) CNN5-SR. (h) ResNet12. (i) **Our method**.

are very difficult to distinguish in SAR images. The result of ResNet12 is shown in Fig. 9(h), where the classification of water is improved, but this method has a certain degree of misclassification for residential. Our approach realizes the best performances for SAR image scene classification, as shown in Fig. 9(i), which obtains consistent regions in the homogeneous subspace, achieves accurate classification in the hybrid subspace, and keeps details in the structural subspace due to adopting the adaptive scene classification method.

The hybrid pixel subspace in the Pyram image has a great intraclass difference and interclass similarity, and the structural pixel subspace has no obvious visual semantic features. Therefore, the labeling of samples and locating structures become challenging problems. The main purpose of designing the fused region map is to make the annotation more convincing. Under the guidance of the fused regional map [see Fig. 10(d)], we label three scenes for the hybrid pixel subspace and present three enlarged image blocks, as shown in Fig. 11(a)–(c), to show the differences between the three scenes. Classification results of the Pyram image are shown in Fig. 10(e)–(i), where different colors denote sparse residential (red), middle residential (green), dense residential (blue), and idle land (yellow). It can be seen that the GLCM-MLR method [see Fig. 10(e)] fails to distinguish the three residential scenes, especially the sparse residential scene, and there are serious misclassifications. The result of Gabor-SVM [see Fig. 10(f)] is better than that of GLCM-MLR, whose classification performance is improved to some extent. The regional consistency of CNN5-SR [see Fig. 10(g)] is superior to the results of Gabor-SVM and

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(a) (b) (c)

Fig. 11. Three enlarged image blocks for the hybrid pixel subspace. (a) Sparse residential. (b) Middle residential. (c) Dense residential.

GLCM-MLR. The improvement of classification accuracy is attributed to the fact that CNN5-SR can realize end-to-end feature extraction and classification. Compared with CNN5-SR, the classification result of ResNet12 [see Fig. 10(h)] is more accurate because ResNet12 is deeper and can capture more discriminative features than CNN5. We can see from the experimental results that the performance of the proposed method [see Fig. 10(i)] is optimal. On the basis of the adaptive classification method, it obtains the best regional consistency and classification accuracy. In addition, we locate some line targets in the structural pixel space, marked in black color. Some isolated targets, such as the pyramids, have not been located effectively.

C. Quantitative Analysis of SAR Images

To calculate quantitative indexes, such as overall classification accuracy (OA) and the kappa coefficient, we select five typical scenes from the hybrid pixel subspace and four scenes from the homogeneous pixel subspace of SAR images and construct a nine-scene SAR image data set with a size of 200×200, as shown in Fig. 12. To verify the performance of the proposed method more accurately, the first five images use the Ridgelet-ResNet12, and the rest utilize the Ridgelet-CNN5-SR for classification. Then, we calculate their average indexes. We randomly sample 1000 24×24 patches for each scene, where 50, 100, 200, 400, and 600 patches are the training samples, and the rest are the testing samples. We calculate the OA and kappa coefficient 20 times per method, and the average results are shown in Tables II and III.

Table II shows the OAs of the five methods using different numbers of training samples. It is obvious that the OAs of GLCM-MLR are the lowest in all cases, and the results of Gabor-SVM are higher than those of GLCM-MLR. We can also see from Table II that the CNN-based methods outperform the approaches with handcrafted features. Our method achieves the highest accuracy on the classification of nine-class scenes and is significantly superior to the other methods, especially when the training samples are very small. This is because the ridgelet kernels have fewer network parameters and can successfully avoid the overfitting issue associated with limited labeled samples.

Table III presents the results of the Kappa coefficient in different methods. The Kappa coefficients of our method are the highest under different numbers of classes, which further shows that the proposed algorithm performs well on very limited labeled samples and is superior to the comparative

TABLE II

OA (%) ON NINE-SCENE DATA SET OBTAINED WITH DIFFERENT TRAINING SAMPLES PER CLASS

Methods	50/class	100/class	200/class	400/class	600/class
GLCM-MLR	67.91	73.38	81.41	85.25	89.03
Gabor-SVM	73.34	76.80	83.53	87.43	90.89
CNN5-SR	80.51	86.35	90.52	93.61	94.96
ResNet12	81.28	87.12	91.94	93.85	94.57
Our method	**86.03**	**91.29**	**94.71**	**94.98**	**95.02**

TABLE III

KAPPA COEFFICIENT ON NINE-SCENE DATA SET OBTAINED WITH DIFFERENT TRAINING SAMPLES PER CLASS

Methods	50/class	100/class	200/class	400/class	600/class
GLCM-MLR	0.5328	0.6730	0.7428	0.7632	0.7964
Gabor-SVM	0.6597	0.7329	0.7709	0.7996	0.8351
CNN5-SR	0.7431	0.7963	0.8519	0.8657	0.8902
ResNet12	0.7639	0.8121	0.8657	0.8721	0.8836
Our method	**0.8008**	**0.8652**	**0.8972**	**0.9057**	**0.9074**

methods in all cases. The excellent performance indicates that the Ridgelet-Net with speckle reduction regularization is an effective way to deal with small-sample learning problems.

To further verify and show the proposed method's effectiveness, we construct a synthetic SAR image with size 600×600 (denoted as SYN1) to perform the compared experiment. It contains five scenes coming from real SAR images, as shown in Fig. 13(a). Fig. 13(b) shows the corresponding ground-truth map. The image is relatively small, we sample 200 patches with size 12×12 for each scene as training samples, and other parameters are consistent with the default parameters in this article. The compared experiments include GLCM-MLR, Gabor-SVM, CNN5-SR, and ResNet12. We calculate the classification maps and quantitative indexes (accuracy per category, OA, and kappa) of each method, and the experimental results are shown in Fig. 13 and Table IV.

From the results, we can see that the regional consistency and accuracy of edges in GLCM-MLR [see Fig. 13(c)] and Gabor-SVM [see Fig. 13(d)] are inferior to other methods. The classification performances of CNN5-SR [see Fig. 13(e)] and ResNet12 [see Fig. 13(f)] are improved greatly, especially the idle land, farmland, and grassland scenes. Our method [see Fig. 13(g)] achieves the optimal classification result, including the boundary precision and semantic consistency, which proves the effectiveness of the proposed method under the condition of very limited training samples.

In addition, the numerical results are listed in Table IV. Our method obtains a higher classification accuracy on the idle land, farmland, and forest scenes than the compared methods and yields the highest OA and kappa. The reason is that the Ridgelet-Net with regularization is trained end-to-end under the guidance of labels and can adaptively approximate the distribution parameters of SAR images.

D. Parameter Analysis

There are three parameters that deserve to be analyzed and discussed in detail: patch size, number of filters, and hyperparameter. The size of image patches is set as 8, 16, 24, 32, 64,

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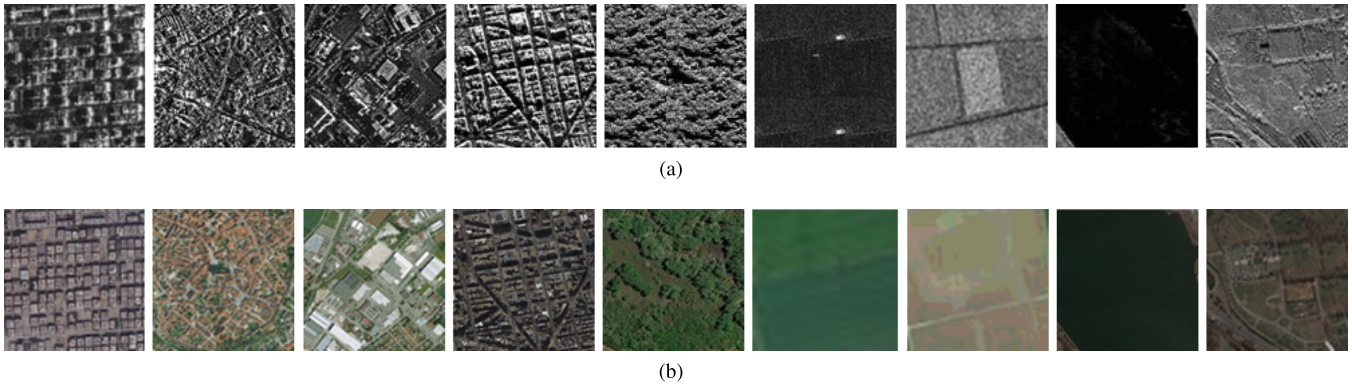


Fig. 12. Nine-scene data set (residential, dense residential, industrial, commercial, forest, grassland, agricultural, water, and idle land). (a) Original SAR image patches. (b) Corresponding optical image patches.

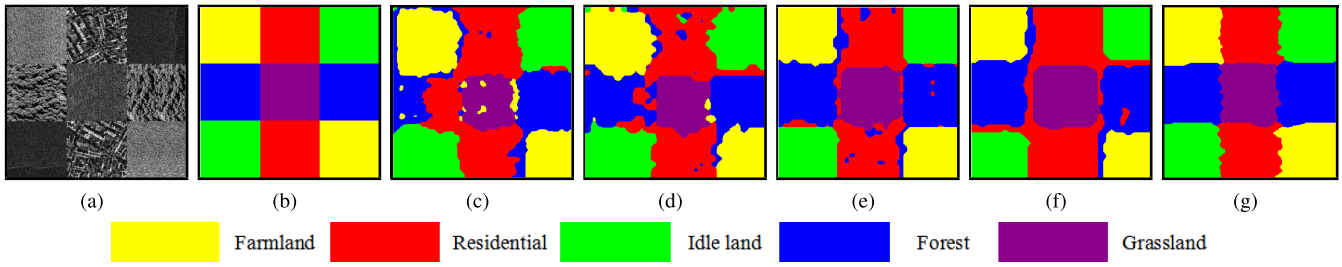


Fig. 13. Classification maps of SYN1. (a) SYN1 (600 × 600). (b) Ground truth. (c) GLCM-MLR. (d) Gabor-SVM. (e) CNN5-SR. (f) ResNet12. (g) **Our method**.

TABLE IV
KAPPA COEFFICIENT ON NINE-SCENE DATA SET OBTAINED WITH DIFFERENT TRAINING SAMPLES PER CLASS

Method	**Idle land** (yellow)	**Residential** (red)	**Farmland** (green)	**Forest** (blue)	**Grassland** (purple)	**OA**	**Kappa**
GLCM-MLR	76.43	87.04	87.51	73.22	85.87	82.01	0.7678
Gabor-SVM	78.32	86.75	88.51	80.39	87.65	84.32	0.8082
CNN5-SR	89.71	86.12	93.61	90.27	92.69	90.48	0.5897
ResNet12	88.23	**96.92**	90.88	93.85	**97.22**	93.42	0.8946
Our method	**97.60**	93.28	**95.37**	**96.62**	94.19	**95.41**	**0.9218**

TABLE V
AA (%), OA (%), AND KAPPA COEFFICIENT OF SYN2

Method	**(Pond)** (yellow)	**Farmland** (red)	**Water** (green)	**Mountain** (blue)	**AA**	**OA**	**Kappa**
CNN5	85.76	95.58	93.44	97.11	92.97	92.69	0.8908
Ridgelet-CNN5	87.41	96.26	95.58	98.04	94.32	94.11	0.9159
Ridgelet-CNN5-L2	89.15	96.83	94.72	98.39	94.77	94.32	0.9216
Ridgelet-CNN5-KL	90.92	97.26	96.29	98.71	95.8	95.68	0.9322
Our method	**91.53**	**98.45**	**97.04**	**99.26**	**96.57**	**96.45**	**0.9467**

the number of filters in the first convolutional layer is set as $16(N_\alpha = 2, N_\theta = 4, N_\beta = 2)$, $32(N_\alpha = 4, N_\theta = 4, N_\beta = 2)$, $64(N_\alpha = 4, N_\theta = 8, N_\beta = 2)$, and $128(N_\alpha = 4, N_\theta = 8, N_\beta = 4)$, and the hyperparameter λ takes values from 0.1 to 1 with a step size of 0.1. We use the nine-scene data set with 200 training samples to calculate the OA under various parameter values. Each parameter is calculated ten times, and the influence of the parameters on OA is plotted in Fig. 14. Finally, we investigate the performance of the two different initialization methods: uniform initialization (Ridgelet-uniform) and sketch statistics-based initialization (Ridgelet-statistic). At the same time, the proposed methods are compared with common CNNs (CNN5). We plot the loss and accuracy curves of 50 epochs, as shown in Fig. 15. Fig. 14(a) shows the OAs for different patch sizes. It can be seen that, when the size of the patch is 16, the OA is the highest. Thereafter,

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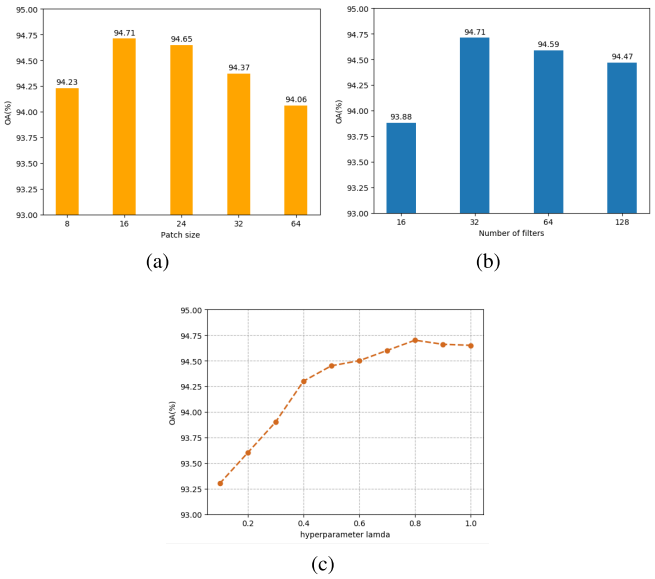


Fig. 14. Influence of different parameters on OAs. (a) Size of image patches. (b) Number of filters. (c) Hyperparameter λ.

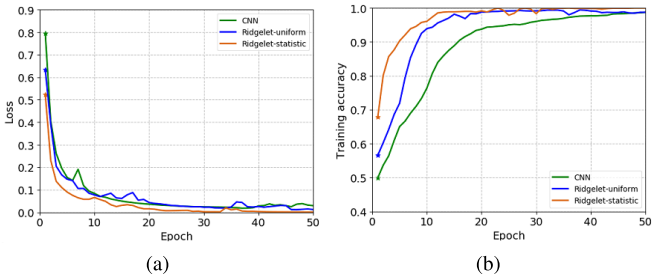


Fig. 15. Convergence rate analysis of different initialization methods. (a) Loss function curves. (b) Training accuracy curves.

the accuracy declines slowly with increasing patch size. This indicates that the patch size for 16 or 24 is enough to represent the features of SAR images, so we set the patch size as 16 in the homogeneous pixel subspace and 24 in the hybrid pixel subspace. Fig. 14(b) illustrates the OAs for different filter numbers. Noticeably, when the number of filters is 16, the accuracy is the lowest. As the filter number increases, our method obtains the highest OA of 94.71% and then decreases. The larger number will increase the computational complexity greatly, so we select the filter number in the first convolutional layer to be 32, and other layers still follow the rules in Section IV-A. The OAs with different hyperparameters λ are presented in Fig. 14(c). We can see that the OA curve is rising significantly with the increase in parameter values, reaching a peak at 0.8 and then decreasing slightly. Thus, the hyperparameter λ is set as 0.8 in our experiments. Fig. 15 shows the compared results of loss and accuracy curves of the three methods. It can be seen that the Ridgelet-uniform method exhibits a smaller initial loss, a higher training accuracy, and a faster convergence speed than the CNN5 method, while the Ridgelet-statistic method has the lowest loss and highest training accuracy, and the convergence rate is also the fastest. In addition, the initial loss of Ridgelet-statistic is smaller than that of the Ridgelet-uniform and CNN5, and the initial

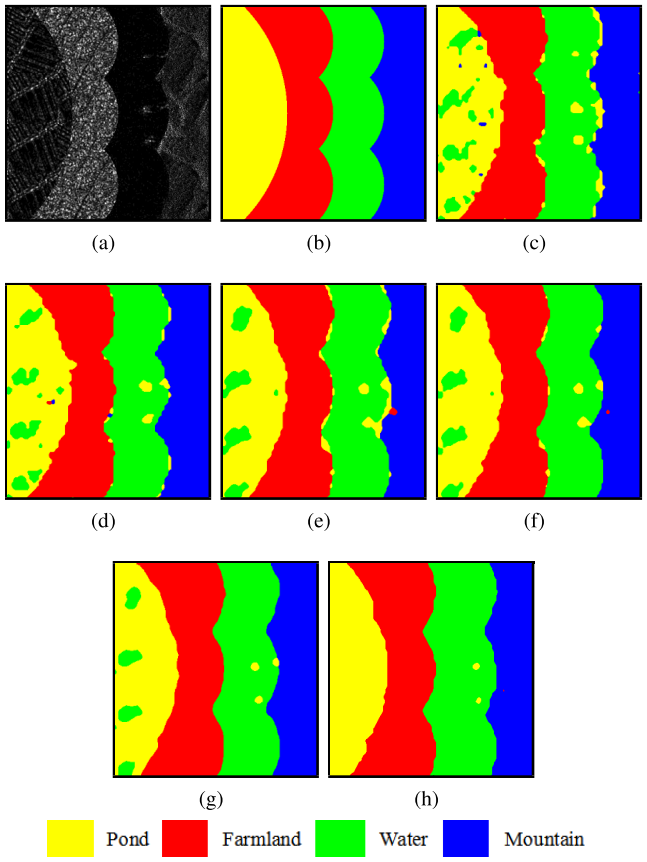


Fig. 16. Classification maps of SYN2. (a) SYN2 image (420 × 400). (b) Ground truth. (c) CNN5. (d) Ridgelet-CNN5. (e) Ridgelet-CNN5-L2. (f) Ridgelet-CNN5-KL. (g) **Our method**. (h) **Our method** (600 training samples).

training accuracy of the Ridgelet-statistic is higher than that of the Ridgelet-uniform and CNN5 due to the introduction of prior information of SAR images. All these results verify the effectiveness of the proposed methods.

E. Ablation Study

The regularization is mainly used in the homogeneous pixel subspace, so we construct another synthetic image (denoted as SYN2) with four scenes (pond, farmland, water, and mountain) to perform the ablation study, as shown in Fig. 16(a). The ground-truth map is shown in Fig. 16(b). We design five compared experiments, including CNN5, Ridgelet-CNN5, Ridgelet-CNN5 with the sparse reconstruction term (Ridgelet-CNN5-L2), Ridgelet-CNN5 with the KL distance restraint (Ridgelet-CNN5-KL), and Ridgelet-CNN5 with the speckle reduction regularization (our method), to qualitatively and quantitatively evaluate the effectiveness of the proposed method. We calculate the classification accuracy of each category and three quantitative indexes: average accuracy (AA), OA, and kappa coefficient. All the results are shown in Fig. 16 and Table V.

The classification results of the ablation study are shown in Fig. 16. It can be seen that the regional consistency and edge accuracy of the CNN5 method are the worst, as shown in Fig. 16(c). The classification map of Ridgelet-CNN5 is shown in Fig. 16(d) and is better than the result of CNN5;

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in particular, the edge accuracy is greatly improved. This is because the ridgelet filters exhibit high sensitivity to linear singularities. When introducing the L_2-norm and KL distance, as shown in Fig. 16(e) and (f), their regional consistency is improved compared with that of Ridgelet-CNN5. Our method achieves the best classification accuracy with the combining of Ridgelet-Nets and speckle reduction regularization. Especially, when the training samples are increased to 600, the classification accuracy is further improved, as shown in Fig. 16(h).

Having the ground-truth information, we also calculate the numerical results of AA, OA, and kappa coefficient, as shown in Table V. From the results, we can see that the AA, OA, and kappa of the CNN5 method are the lowest, and the Ridgelet-CNN5 method obtains higher results than that of CNN5. Moreover, the two regularization terms further improve the AA, OA, and kappa coefficient indexes in Ridgelet-CNN5-L2 and Ridgelet-CNN5-KL, and our method acquires the best results compared with the other methods. The ablation study demonstrates that the two parts of the proposed method are effective in SAR image scene classification.

V. Conclusion

This article considers the quantity and quality of training samples together and proposes Ridgelet-Nets with speckle reduction regularization for SAR image scene classification. On the one hand, the designed ridgelet kernels are general and can be easily embedded in any deep neural network to compose Ridgelet-Net, which has fewer parameters, faster convergence speed, and more powerful representation than common CNNs. On the other hand, we construct a new objective function with speckle reduction regularization to optimize the Ridgelet-Net. The proposed method contends with the low quality of training samples and improves the classification performance of SAR images. Finally, we employ an adaptive SAR image classification framework for different pixel subspaces based on the extended hierarchical visual semantic model. The experiments show promising results, especially when fewer labeled samples are available. However, the location of the structural pixel subspace is not very accurate. How to design effective location rules is a problem worth studying. In addition, constructing novel multiscale geometric neural networks is also an interesting research direction.

References

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[2] H. Yu, L. Jiao, and F. Liu, "CRIM-FCHO: SAR image two-stage segmentation with multifeature ensemble," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2400–2423, Apr. 2016.

[3] C. O. Dumitru, G. Schwarz, and M. Datcu, "SAR image land cover datasets for classification benchmarking of temporal changes," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 5, pp. 1571–1592, May 2018.

[4] Z. Huang, Z. Pan, and B. Lei, "What, where, and how to transfer in SAR target recognition based on deep CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2324–2336, Apr. 2020.

[5] R. Huber, "Scene classification of SAR images acquired from antiparallel tracks using evidential and rule-based fusion," *Image Vis. Comput.*, vol. 19, no. 13, pp. 1001–1010, Nov. 2001.

[6] S. Yang, M. Wang, H. Long, and Z. Liu, "Sparse robust filters for scene classification of synthetic aperture radar (SAR) images," *Neurocomputing*, vol. 184, pp. 91–98, Apr. 2016.

[7] J.-S. Lee, J.-S. Lee, J.-H. Wen, T. L. Ainsworth, K.-S. Chen, and A. J. Chen, "Improved sigma filter for speckle filtering of SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 1, pp. 202–213, Jan. 2009.

[8] F. Dell'Acqua, P. Gamba, and G. Trianni, "Semi-automatic choice of scale-dependent features for satellite SAR image classification," *Pattern Recognit. Lett.*, vol. 27, no. 4, pp. 244–251, Mar. 2006.

[9] D. Borghys, Y. Yvinec, C. Perneel, A. Pizurica, and W. Philips, "Supervised feature-based classification of multi-channel SAR images," *Pattern Recognit. Lett.*, vol. 27, no. 4, pp. 252–258, Mar. 2006.

[10] S. Guolei and H. Wei, "Overview of SAR image segmentation algorithm," *Comput. Era*, vol. 5, pp. 1–5, May 2017.

[11] V. V. Chamundeeswari, D. Singh, and K. Singh, "An adaptive method with integration of multi-wavelet based features for unsupervised classification of SAR images," *J. Geophys. Eng.*, vol. 4, no. 4, pp. 384–393, Oct. 2007.

[12] W. Li and Q. Du, "Gabor-filtering-based nearest regularized subspace for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1012–1022, Apr. 2014.

[13] N. Zheng and H. F. Tan, "SAR image classification based on brushlet and self-adaptive ridgelet neural network," *Appl. Mech. Mater.*, vols. 347–350, pp. 3024–3028, Aug. 2013.

[14] Q. Yu, C. Moloney, and F. M. Williams, "SAR sea-ice texture classification using discrete wavelet transform based methods," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jun. 2002, pp. 3041–3043.

[15] Z. Liu, "Minimum distance texture classification of SAR images in contourlet domain," in *Proc. Int. Conf. Comput. Sci. Softw. Eng.*, 2008, pp. 834–837.

[16] Y.-Q. Li, M.-Y. He, and C.-X. Li, "An approximate finite ridgelet transform algorithm for SAR image segmentation," in *Proc. 8th Int. Conf. Signal Process.*, Nov. 2006, pp. 13–15.

[17] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.

[18] A. Krizhevsky, I. Sutskever, and G. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, no. 2, 2012, pp. 84–90.

[19] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[20] H. Furukawa, "Deep learning for target classification from SAR imagery: Data augmentation and translation invariance," IEICE, Nagasaki, Japan, Tech. Rep. SANE2017-30, Aug. 2017.

[21] W. Lei, X. Xin, D. Hao, G. Rong, and F. Pu, "Multi-pixel simultaneous classification of PolSAR image using convolutional neural networks," *Sensors*, vol. 18, no. 3, pp. 769–781, Mar. 2018.

[22] Z. Zhao, L. Jiao, J. Zhao, J. Gu, and J. Zhao, "Discriminant deep belief network for high-resolution SAR image classification," *Pattern Recognit.*, vol. 61, pp. 686–701, Jan. 2017.

[23] J. Zhao, W. Guo, S. Cui, Z. Zhang, and W. Yu, "Convolutional neural network for SAR image classification at patch level," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 945–948.

[24] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.

[25] Z. Huang, Z. Pan, and B. Lei, "Transfer learning with deep convolutional neural network for SAR target classification with limited labeled data," *Remote Sens.*, vol. 9, no. 9, pp. 907–928, Aug. 2017.

[26] J. Geng, H. Wang, J. Fan, and X. Ma, "Deep supervised and contractive neural network for SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2442–2459, Apr. 2017.

[27] E. J. Candes, "Ridgelets: Theory and application," Ph.D. dissertation, Dept. Statist., Stanford Univ., Stanford, CA, USA, 1998.

[28] M. Mejail, A. C. Frery, J. Jacobo-Berlles, and O. H. Bustos, "Approximation of distributions for SAR images: Proposal, evaluation and practical consequences," *Latin Amer. Appl. Res.*, vol. 31, pp. 83–92, Oct. 2002.

[29] M. Yu, G. Dong, H. Fan, and G. Kuang, "SAR target recognition via local sparse representation of multi-manifold regularized low-rank approximation," *Remote Sens.*, vol. 10, no. 2, pp. 211–233, Feb. 2018.

[30] A. D. C. Nascimento, R. J. Cintra, and A. C. Frery, "Hypothesis testing in speckled data with stochastic distances," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 1, pp. 373–385, Jan. 2010.

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[31] J. Wu, F. Liu, L. Jiao, X. Zhang, H. Hao, and S. Wang, "Local maximal homogeneous region search for SAR speckle reduction with sketch-based geometrical kernel function," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 9, pp. 5751–5764, Sep. 2014.
[32] F. Liu *et al.*, "Hierarchical semantic model and scattering mechanism based PolSAR image classification," *Pattern Recognit.*, vol. 59, pp. 325–342, Nov. 2016.
[33] F. Liu *et al.*, "SAR image segmentation based on hierarchical visual semantic and adaptive neighborhood multinomial latent model," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 4287–4301, Jul. 2016.
[34] S. Yang, W. Min, L. Zhao, and Z. Wang, "Image noise reduction via geometric multiscale ridgelet support vector transform and dictionary learning," *IEEE Trans. Image Process.*, vol. 22, no. 11, pp. 4161–4169, Nov. 2013.
[35] Y. Duan *et al.*, "Adaptive hierarchical multinomial latent model with hybrid kernel function for SAR image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5997–6015, Oct. 2018.
[36] J. M. Fadili and J. L. Starck, "Curvelets and ridgelets," in *Encyclopedia of Complexity and Systems Science*. New York, NY, USA: Springer, Nov. 2012, pp. 1718–1738.
[37] S. Chitroub, A. Houacine, and B. Sansal, "Statistical characterisation and modelling of SAR images," *Signal Process.*, vol. 82, no. 1, pp. 69–92, Jan. 2002.
[38] A. C. Frery, C. da Costa Freitas Yanasse, and S. J. S. Sant'Anna, "Alternative distributions for the multiplicative model in SAR images," in *Proc. Int. Geosci. Remote Sens. Symp. (IGARSS) Quant. Remote Sens. Sci. Appl.*, vol. 1, no. 1, 2002, pp. 169–171.
[39] M. E. Mejail, J. C. Jacobo-Berlles, A. C. Frery, and O. H. Bustos, "Classification of SAR images using a general and tractable multiplicative model," *Int. J. Remote Sens.*, vol. 24, no. 18, pp. 3565–3582, Jan. 2003.
[40] R. C. P. Marques, F. N. Medeiros, and J. S. Nobre, "SAR image segmentation based on level set approach and G0 model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 10, pp. 2046–2057, Oct. 2012.
[41] M. N. Do and M. Vetterli, "The finite ridgelet transform for image representation," *IEEE Trans. Image Process.*, vol. 12, no. 1, pp. 16–28, Jan. 2003.
[42] C. Liu, J. Li, L. He, A. J. Plaza, S. Li, and B. Li, "Naive Gabor networks for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 376–390, Jan. 2021.
[43] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. IEEE Int. Conf. ICLR*, May 2014, pp. 1–14.
[44] K. He and J. Sun, "Convolutional neural networks at constrained time cost," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 5353–5360.



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